	Ans: $(1.1 \pm 0.1) \times 10^{-3}$ kg m^{-1} s^{-1}
2	
(a)	Electric field strength at a point is defined as the electric force per unit charge
	on a positive charge placed at that point.
(b)(i)	$E = \frac{1}{4\pi\varepsilon_o}\frac{Q}{r^2}$ $= \frac{1}{4\pi\varepsilon_o}\frac{\left(1.6\times10^{-19}\right)}{\left(2.8\times10^{-10}\right)^2}$
	$E = 1.8\times10^{10}$ N C^{-1}
(b)(ii)	$F = qE$ $= \left(1.6\times10^{-19}\right)\left(1.835\times10^{10}\right)$ $= 2.9\times10^{-9}$ N
(c)(i)	Nearest atoms are negatively charged and will exert an attractive force on the ion.
	Next nearest atoms are positively charged and will exert an repulsive force on the ion.
	However, due to the symmetry, for every force there is another force in the opposite direction of the same magnitude. Hence the resultant force is zero.

(c)(ii)	The central ion will vibrate about its equilibrium position.
(c)(iii)	Lowest electrical potential energy.
(d)	As the equilibrium distance for sodium chloride is shorter, the electrical potential energy is more negative.
	Since the change in potential energy is higher, a higher temperature is needed to separate the ions when melting.
3	
(a)(i)1	*m* is the mass of a single gas molecule